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Subject : AWS

Problem Statement No : 90

Problem Statement Title : 90. eSportArena Competitive Gaming Cloud

Sem : 4 , **Sprint :** 02

Industry: Esports & Competitive Gaming

Github Repo Link :

Documentation Link :

1. Abstract

eSportArena is a cloud-based gaming platform infrastructure deployed on Amazon Web Services (AWS). The project demonstrates the implementation of core AWS services including VPC, Public Subnets, Security Groups, EC2 Instance, Ubuntu Server, and Nginx Web Server.

The objective is to create a scalable, secure, and highly available cloud environment capable of hosting web applications. The website is successfully deployed on an EC2 instance and accessed through a public IP address.

2. Objectives

- Deploy a website on AWS Cloud.
- Configure networking using VPC and Subnets.
- Launch and manage an EC2 instance.
- Install Ubuntu and Nginx Web Server.
- Understand cloud infrastructure deployment.
- Demonstrate secure and scalable cloud architecture.

3. Technologies Used

Technology	Purpose
AWS EC2	Virtual Server
AWS VPC	Network Isolation
Public subnet	Internet Access
Security Group	Firewall Rules
Ubuntu 24.04	Operating System
Nginx	Web Server
Draw.io	Architecture Diagram
HTML/CSS	Landing page

4. Folder Structure

eSportArena-AWS-Project/

- | |— README.md
- |— Architecture.md
- |— Project_Report.docx
- |— Project_Presentation.pptx
- |— website/
 - | |— index.html
 - | |— styles.css (agar CSS alag file me hai)
- | |— diagrams/
 - | |— aws_architecture.drawio
 - | |— aws_architecture.png
 - | |— aws_architecture.pdf
- | |— screenshots/
 - | |— 01_vpc_created.png

- | |— 02_public_subnets.png
- | |— 03_security_group.png
- | |— 04_ec2_running.png
- | |— 05_ubuntu_terminal.png
- | |— 06_nginx_installed.png
- | |— 07_website_live.png
- | |— 08_architecture_diagram.png
- | |— deployment/
- | |— deployment_steps.md
- | |— nginx_commands.txt
- | |— aws_configuration.txt |
- |— docs/
- |— Abstract.md
- |— Objectives.md
- |— Results.md
- |— Future_Scope.md |— [Conclusion.md](#)

4. Architecture Overview

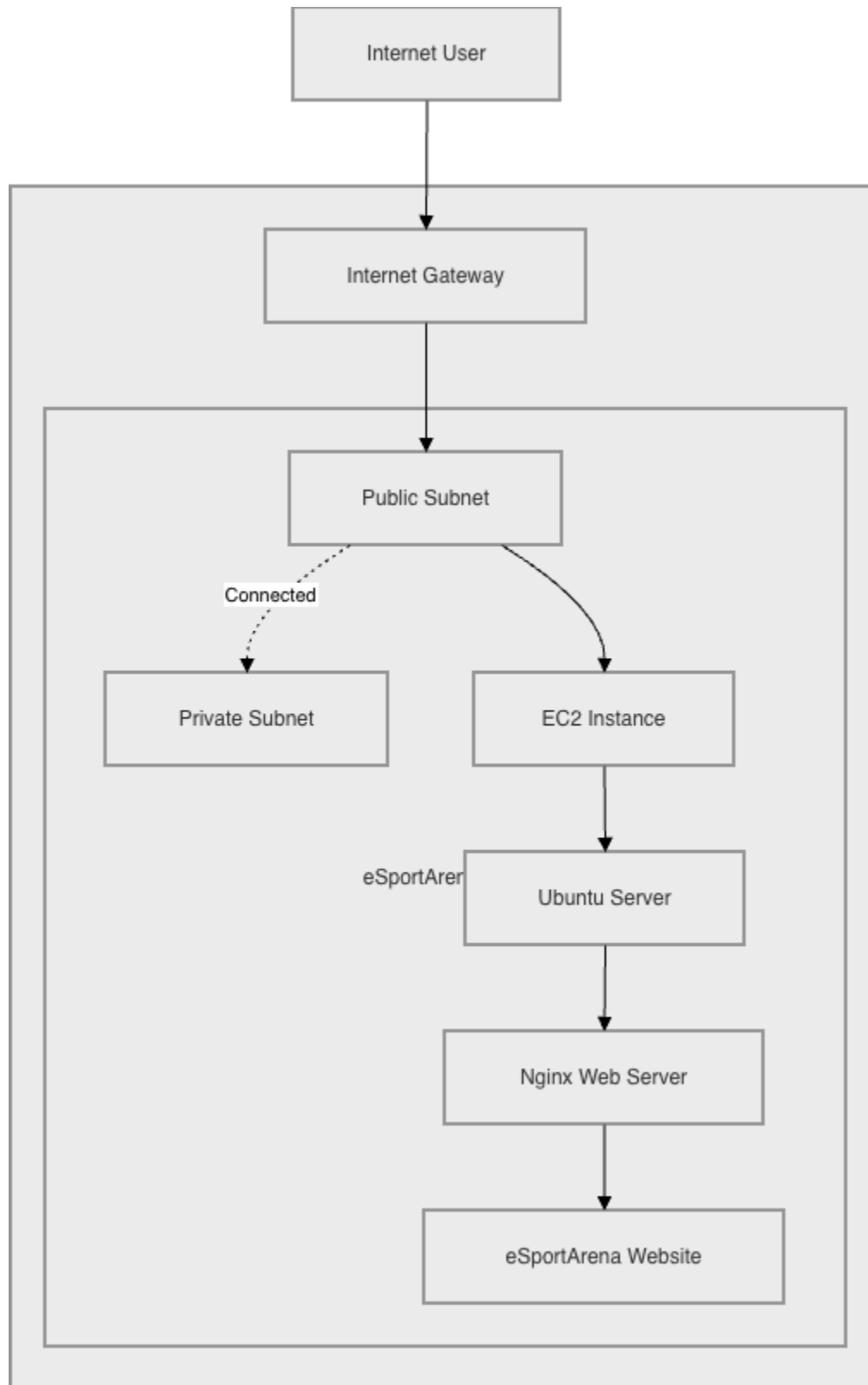


Figure 1: AWS Cloud Architecture of eSportArena Platform

5. Implementation Steps

Step 1: Create VPC

Created a Virtual Private Cloud to isolate network resources.

Step 2: Create Public Subnets

Created two public subnets for resource deployment.

Step 3: Configure Security Groups

Allowed SSH (Port 22) and HTTP (Port 80) traffic.

Step 4: Launch EC2 Instance

Created an Ubuntu-based EC2 instance.

Step 5: Install Nginx

Installed and enabled Nginx web server.

Step 6: Deploy Website

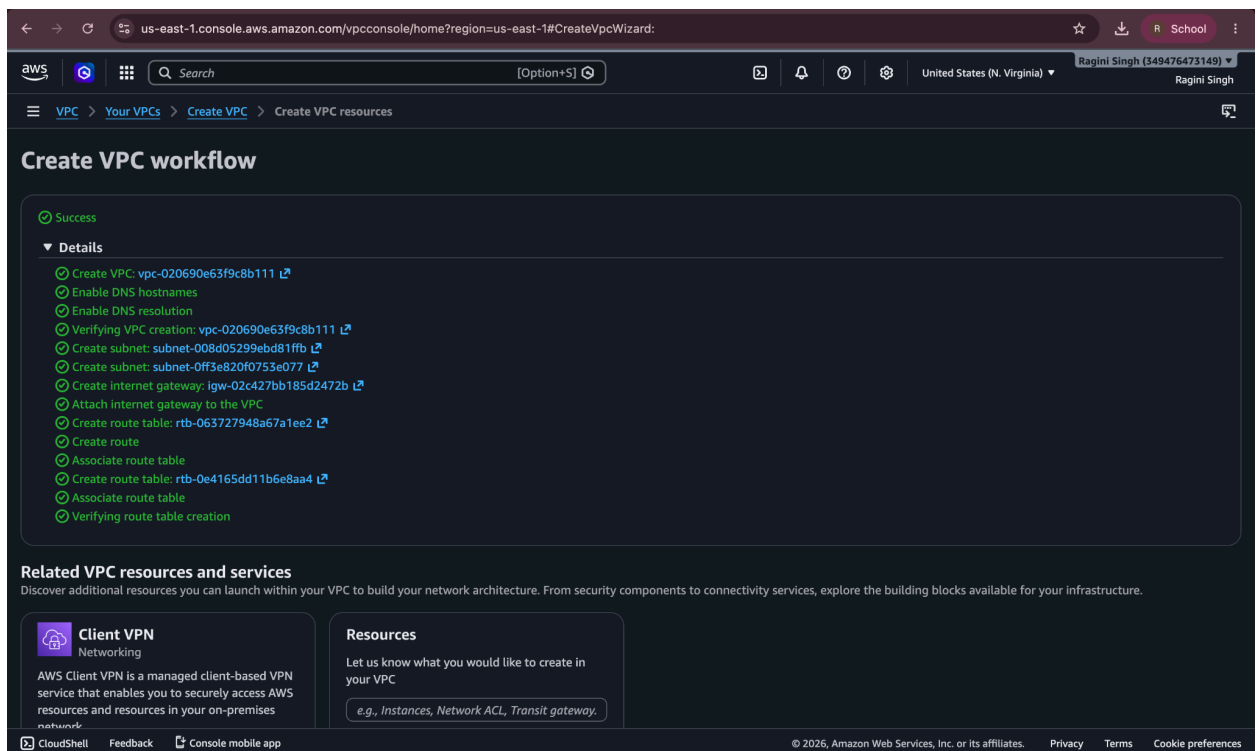
Uploaded a custom HTML page and hosted it using Nginx.

Step 7: Test Deployment

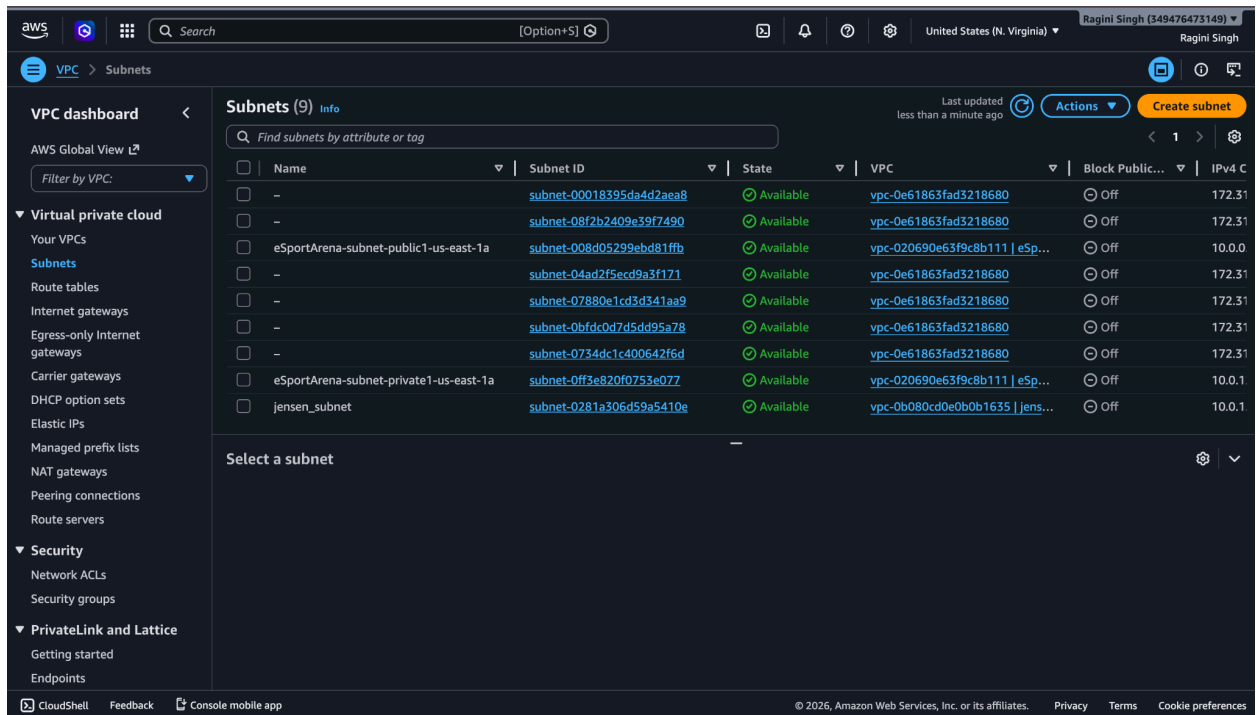
Verified successful deployment using the public IP address.

6. Screenshots

1. VPC Created



2. Subnet Created

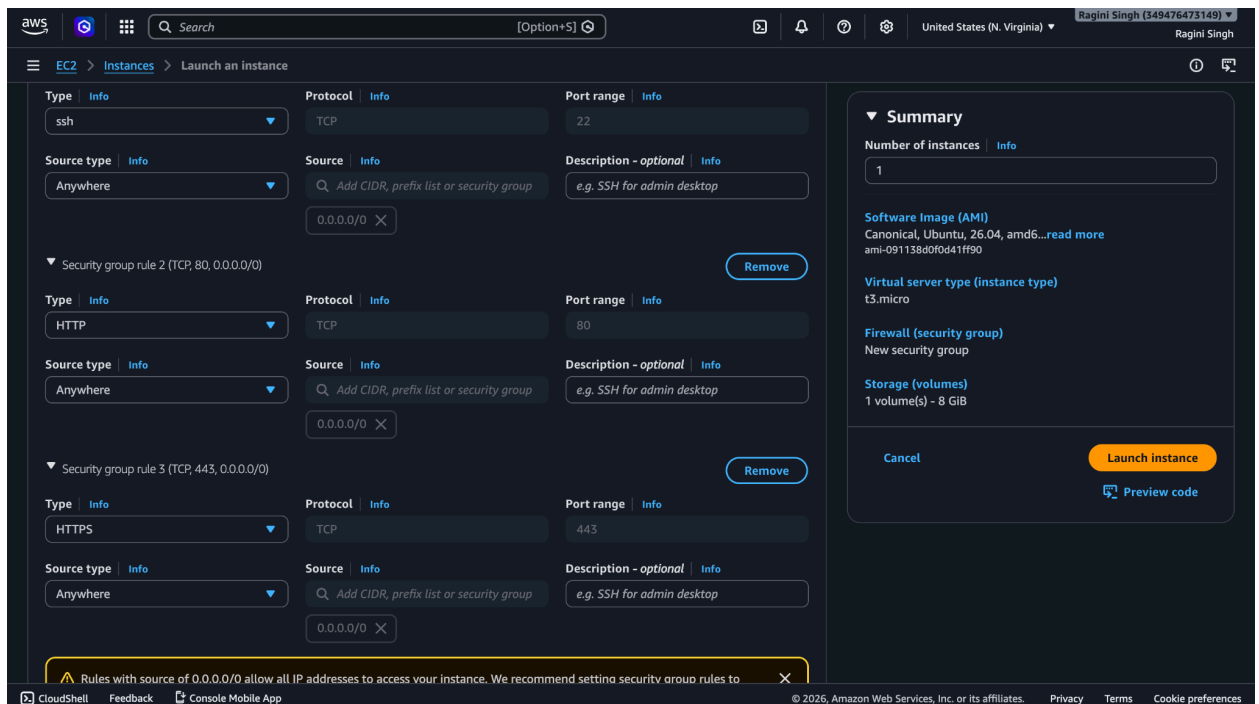


The screenshot shows the AWS Management Console interface for the 'Subnets' page. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Security, and PrivateLink and Lattice. The main content area displays a table of subnets with columns for Name, Subnet ID, State, VPC, Block Public IP, and IPv4 CIDR. The subnets listed are:

Name	Subnet ID	State	VPC	Block Public IP	IPv4 CIDR
-	subnet-00018395da4d2aea8	Available	vpc-0e61863fad3218680	Off	172.31
-	subnet-08f2b2409e39f7490	Available	vpc-0e61863fad3218680	Off	172.31
eSportArena-subnet-public1-us-east-1a	subnet-008d05299ebd81ffb	Available	vpc-020690e63f9c8b111 eSp...	Off	10.0.0
-	subnet-04ad2f5ecd9a3f171	Available	vpc-0e61863fad3218680	Off	172.31
-	subnet-07880e1cd3d341aa9	Available	vpc-0e61863fad3218680	Off	172.31
-	subnet-0bfcd0d7d5dd95a78	Available	vpc-0e61863fad3218680	Off	172.31
-	subnet-0734dc1c400642f6d	Available	vpc-0e61863fad3218680	Off	172.31
eSportArena-subnet-private1-us-east-1a	subnet-0ff3e820f0753e077	Available	vpc-020690e63f9c8b111 eSp...	Off	10.0.1
jensen_subnet	subnet-0281a306d59a5410e	Available	vpc-0b080cd0e0b0b1635 jens...	Off	10.0.1

Below the table, there is a 'Select a subnet' section. The top right corner features a 'Create subnet' button and a 'Last updated' timestamp.

3. Security Group Configuration



The screenshot shows the AWS Management Console interface for the 'Security Group Configuration' page. The left sidebar contains navigation links for EC2, Instances, and Launch an Instance. The main content area displays a table of security group rules with columns for Type, Protocol, Port range, Source type, Source, and Description. The rules listed are:

Type	Protocol	Port range	Source type	Source	Description
ssh	TCP	22	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop
HTTP	TCP	80	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop
HTTPS	TCP	443	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop

Below the table, there is a 'Launch instance' button and a 'Preview code' button. A warning message at the bottom states: 'Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to...'

4. EC2 Instance Running

aws [Search] [Option+S] United States (N. Virginia) Ragini Singh (349476423149) Ragini Singh

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance [\(i-01dcb1deb1a770e59\)](#)

► Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup" < 1 2 3 4 5 6 >

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

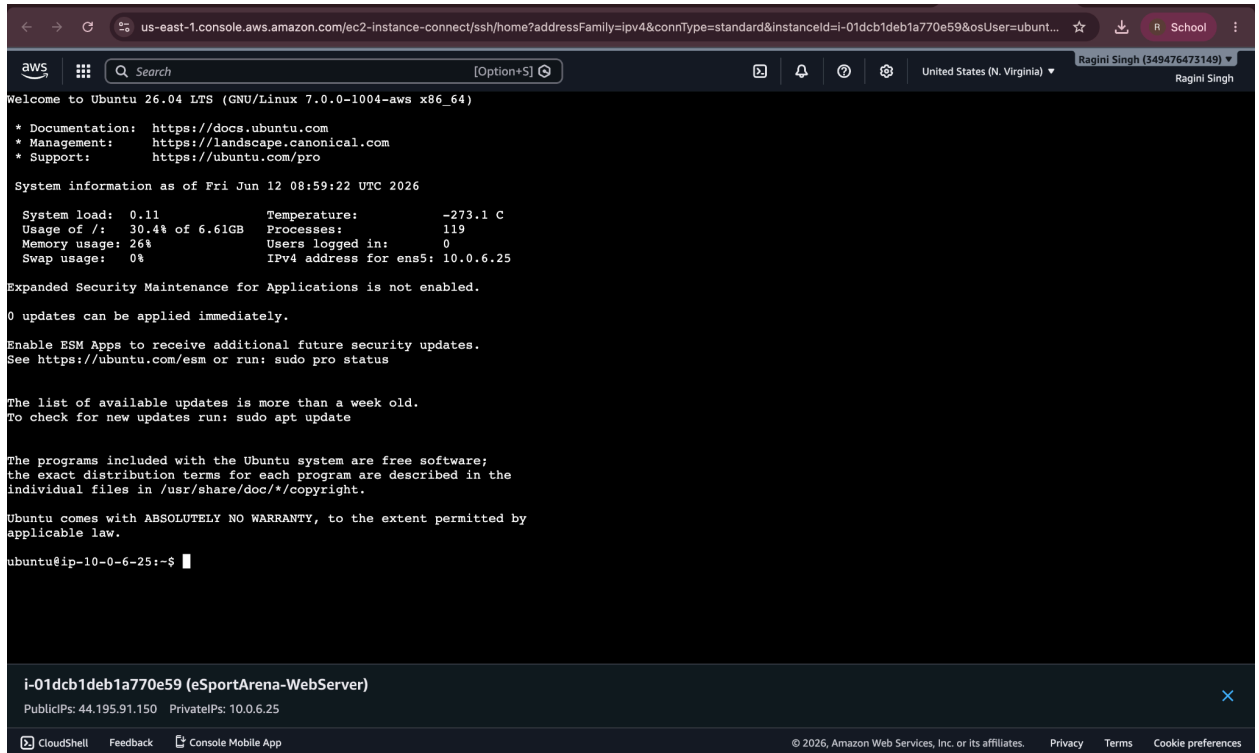
Create Load Balancer
Create an application, network gateway or classic Elastic Load Balancer
[Create Load Balancer](#)

Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
[Manage CloudWatch alarms](#)

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5. Ubuntu Terminal



The screenshot shows an AWS CloudShell terminal window. The browser address bar at the top indicates the connection to an EC2 instance. The terminal itself displays the Ubuntu 26.04 LTS welcome message, including links for documentation, management, and support. It then provides system information as of Friday, June 12, 2026, at 08:59:22 UTC. The system load is 0.11, and the temperature is -273.1 C. Usage of the root filesystem is 30.4% of 6.61GB. There are 119 processes running, and no users are logged in. The IPv4 address for the ens5 interface is 10.0.6.25. The terminal also mentions that Expanded Security Maintenance for Applications is not enabled and that 0 updates can be applied immediately. It advises enabling ESM Apps for future security updates and provides a link to the Ubuntu website. The list of available updates is noted as being more than a week old, and it suggests running 'sudo apt update' to check for new updates. The terminal concludes with a disclaimer that the programs included with the Ubuntu system are free software and that the distribution terms are described in the individual files in /usr/share/doc/*/copyright. It also states that Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law. The prompt is 'ubuntu@ip-10-0-6-25:~\$'. At the bottom of the terminal window, there is a banner for the instance 'i-01dcb1deb1a770e59 (eSportArena-WebServer)' with public and private IP addresses. The footer of the browser window shows the AWS CloudShell logo, a feedback link, a console mobile app link, and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for privacy, terms, and cookie preferences.

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01dcb1deb1a770e59&osUser=ubunt...
aws
Search [Option+S]
United States (N. Virginia) Ragini Singh (349476473149) Ragini Singh

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Jun 12 08:59:22 UTC 2026

System load:  0.11           Temperature:   -273.1 C
Usage of /:    30.4% of 6.61GB Processes:      119
Memory usage: 26%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 10.0.6.25

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-6-25:~$

i-01dcb1deb1a770e59 (eSportArena-WebServer)
PublicIPs: 44.195.91.150 PrivateIPs: 10.0.6.25

CloudShell Feedback Console Mobile App
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```

6. Nginx Installation

```
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.4) ...
 * Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
sudo systemctl start nginxProcessing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-6-25:~$ sudo systemctl start nginx
ubuntu@ip-10-0-6-25:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-6-25:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-12 09:00:38 UTC; 34s ago
     Invocation: 5b5c8b8c3033440db407f123b720aa9a
       Docs: man:nginx(8)
    Main PID: 1998 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 39ms
    CGroup: /system.slice/nginx.service
            └─1998 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─2001 "nginx: worker process"
                 └─2002 "nginx: worker process"

Jun 12 09:00:38 ip-10-0-6-25 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jun 12 09:00:38 ip-10-0-6-25 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-10-0-6-25:~$
```

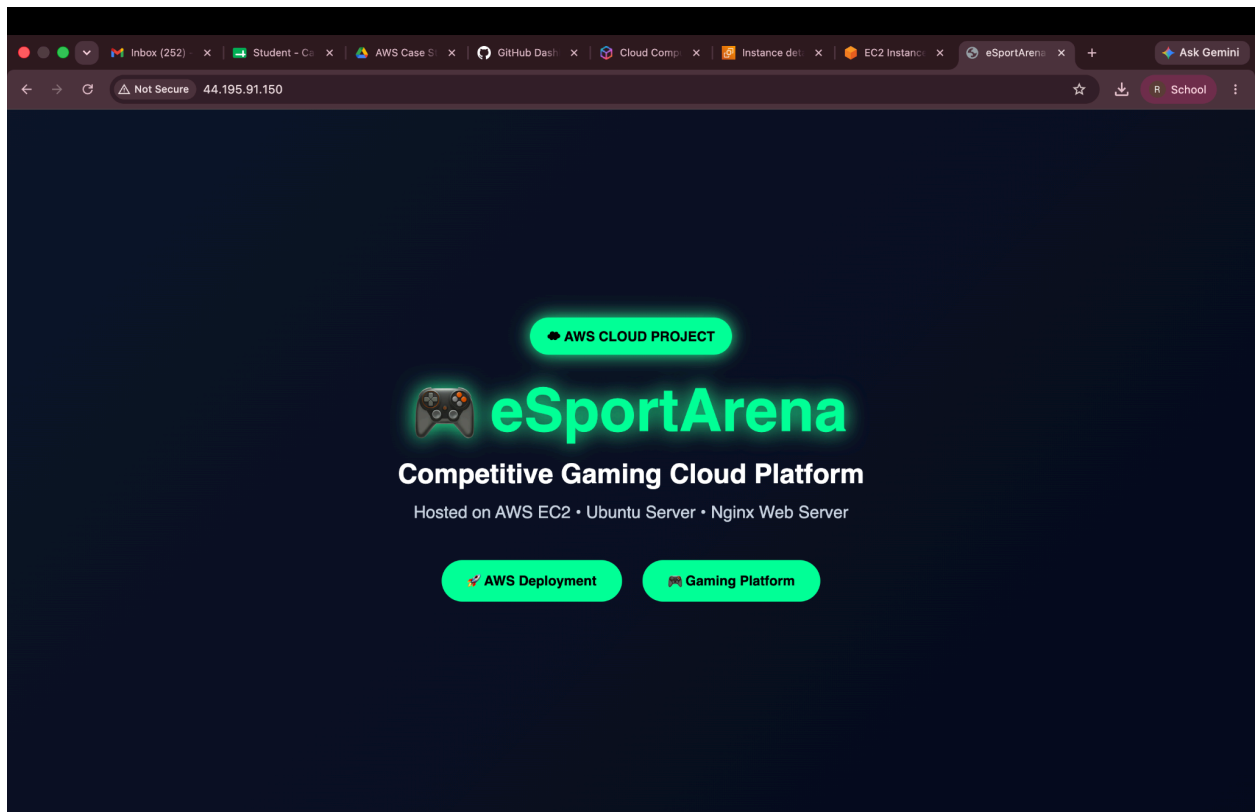
i-01dcb1deb1a770e59 (eSportArena-WebServer)
PublicIPs: 44.195.91.150 PrivateIPs: 10.0.6.25

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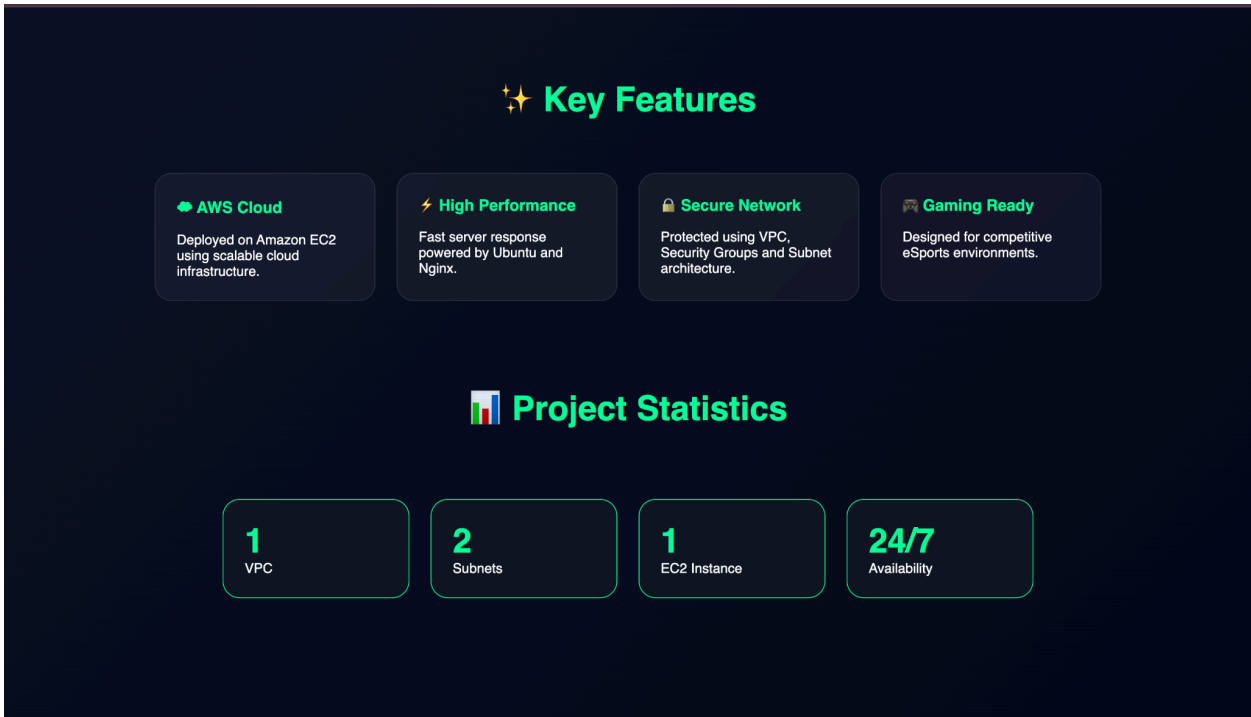
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7. Website Running on Public IP

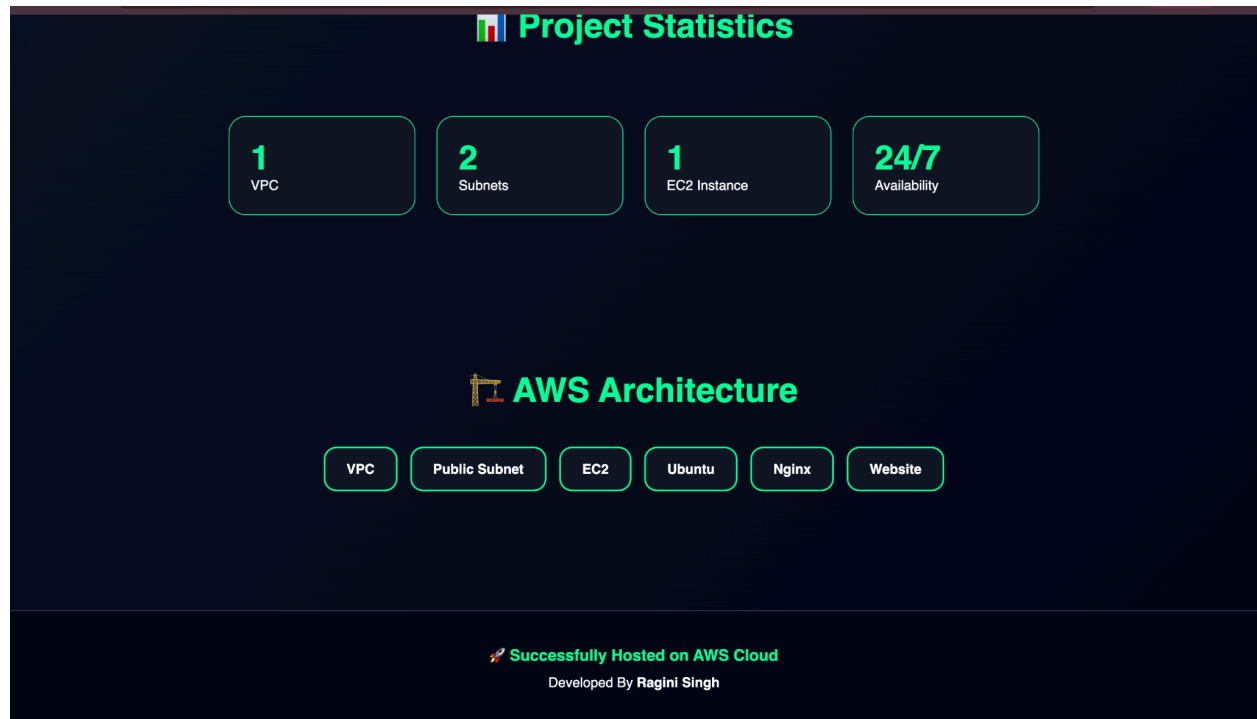
Home page :



Feature:



Final AWS Architecture Diagram



7. Results

The eSportArena web application was successfully deployed on AWS EC2 using Ubuntu and Nginx. The website became accessible through the assigned public IP address. The project demonstrates practical implementation of cloud computing concepts including networking, virtualization, security, and web hosting.

8. Future Scope

- Add Load Balancer

- Implement Auto Scaling
- Configure Route 53 Domain
- Deploy HTTPS using SSL Certificates
- Integrate CloudWatch Monitoring
- Containerize Application using Docker
- Deploy using Kubernetes

9. Conclusion

The project successfully demonstrates the deployment of a cloud-hosted gaming platform infrastructure using AWS services. It provides hands-on experience with EC2, VPC, Security Groups, Ubuntu, and Nginx while showcasing the benefits of scalability, availability, and security offered by cloud computing.

THANKYOUUU!!!!!!